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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2019 Annual Examination

FORM V

MATHEMATICS ADVANCED

Wednesday 4th September 2019

General Instructions

- Writing time — 2 hours
- Write using black pen.
- NESA-approved calculators may be used.

Total — 80 Marks

- All questions may be attempted.

Section I — 8 Marks

- Questions 1–8 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II — 72 Marks

- Questions 9–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Nine.

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Reference Sheet
- Candidature — 177 boys

Examiner

REJ

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which of the following expressions is equivalent to $(3\sqrt{5} - 5)^2$?

- (A) 20
- (B) $70 - 30\sqrt{5}$
- (C) $20 - 30\sqrt{5}$
- (D) 70

QUESTION TWO

It is known that 14 apples and 4 oranges cost \$8, and 5 apples and 4 oranges cost \$4.40. If \$ x represents the cost of an apple and \$ y represents the cost of an orange, which of the following pairs of simultaneous equations could be used to find the cost of each apple and orange?

- (A) $4x + 14y = 8$ and $5x + 4y = 4.4$
- (B) $14x + 4y = 8$ and $5x + 4y = 4.4$
- (C) $14x + 4y = 8$ and $4x + 5y = 4.4$
- (D) $4x + 14y = 8$ and $4x + 5y = 4.4$

QUESTION THREE

Which expression is equivalent to $\log_a 2m - \log_a m$?

- (A) $\log_a m$
- (B) $\frac{\log_a 2m}{\log_a m}$
- (C) $\log_a 2m$
- (D) $\log_a 2$

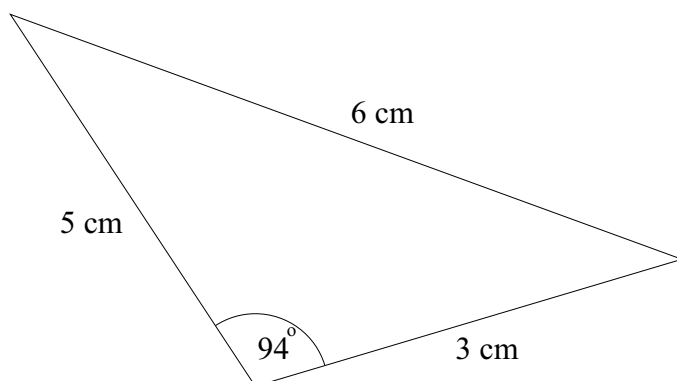
QUESTION FOUR

Which of the following is the derivative of e^{2x} ?

- (A) $2e^{2x-1}$
- (B) $2e^{2x}$
- (C) $2xe^{2x-1}$
- (D) e^{2x}

Examination continues next page ...

QUESTION FIVE



Which expression gives the correct area of the triangle above?

- (A) $\frac{1}{2} \times 5 \times 3 \times \cos 94^\circ$
- (B) $\frac{1}{2} \times 5 \times 6 \times \cos 94^\circ$
- (C) $\frac{1}{2} \times 5 \times 3 \times \sin 94^\circ$
- (D) $\frac{1}{2} \times 5 \times 6 \times \sin 94^\circ$

QUESTION SIX

What is the natural domain of the function $f(x) = \frac{1}{\sqrt{4-x^2}}$?

- (A) $-2 \leq x \leq 2$
- (B) $x \leq -2$ or $x \geq 2$
- (C) $-2 < x < 2$
- (D) $x < -2$ or $x > 2$

QUESTION SEVEN

Each of the following experiments involves two events A and B . In which case are the events A and B independent?

- (A) $P(A|B) = 0.5$ and $P(A) = 0.4$ and $P(B) = 0.5$
- (B) $P(A) = 0.3$ and $P(B) = 0.7$ and $P(A \cap B) = 0.21$
- (C) $P(A|B) = \frac{3}{4}$ and $P(A) = \frac{2}{5}$ and $P(B) = \frac{3}{10}$
- (D) $P(A \cap B) = \frac{1}{5}$ and $P(A) = \frac{1}{5}$ and $P(B) = \frac{2}{3}$

Examination continues overleaf ...

QUESTION EIGHT

Let $f(x) = x^2 + 3x - 4$ and $g(x) = |x|$. What is the correct expression for $f(g(x))$?

(A) $f(g(x)) = \begin{cases} x^2 + 3x - 4, & \text{for } x \geq 0, \\ x^2 - 3x - 4, & \text{for } x < 0 \end{cases}$

(B) $f(g(x)) = \begin{cases} x^2 - 3x - 4, & \text{for } x \geq 0, \\ x^2 + 3x - 4, & \text{for } x < 0 \end{cases}$

(C) $f(g(x)) = \begin{cases} x^2 + 3x - 4, & \text{for } x \leq -4 \text{ or } x \geq 1, \\ -x^2 - 3x + 4, & \text{for } -4 < x < 1 \end{cases}$

(D) $f(g(x)) = \begin{cases} -x^2 - 3x + 4, & \text{for } x \leq -4 \text{ or } x \geq 1, \\ x^2 + 3x - 4, & \text{for } -4 < x < 1 \end{cases}$

————— End of Section I —————

Examination continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION NINE (12 marks) Use a separate writing booklet.**Marks**

(a) Consider the function $f(x) = x^2 - 2x + 3$.

(i) Find the value of $f(2)$.

1

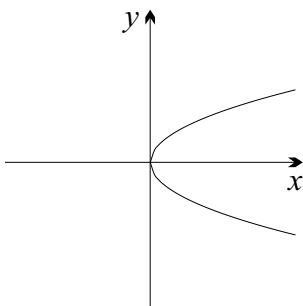
(ii) Find a simplified expression for $f(a + 2)$.

1

(iii) Find a simplified expression for $f(a) + 2$.

1

(b)

1

Classify the graph shown above as one-to-one, many-to-one, one-to-many or many-to-many.

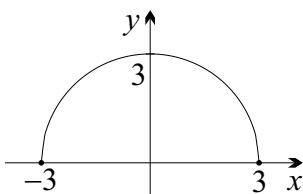
(c) Rationalise the denominator of $\frac{2}{\sqrt{5} - 2}$.

1

(d) Solve $|2x - 1| = 11$.

2

(e)

2

The diagram above shows the graph of the semicircle $y = \sqrt{9 - x^2}$. State the domain and range of this function.

QUESTION NINE (Continued)

(f) A bag contains 2 red discs and 3 white discs. Two discs are drawn from the bag without replacement. What is the probability that both discs are red?

1

(g) Find the exact value of:

(i) $\sin \frac{2\pi}{3}$

1

(ii) $\cos \frac{5\pi}{4}$

1

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

(a) Find the derivative of each of the following functions.

(i) $y = 2x^3$

1

(ii) $y = \frac{1}{x^2}$

1

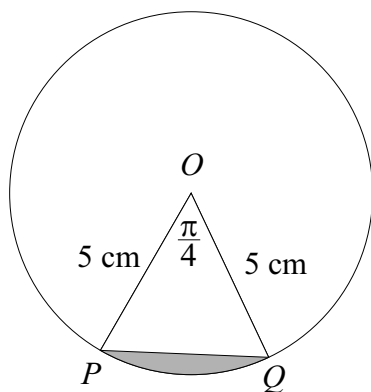
(iii) $y = (3x + 2)^4$

1

(iv) $y = \frac{x^3}{3x + 2}$

2

(b)



The diagram above shows a circle with centre O , radius 5 cm and $\angle POQ = \frac{\pi}{4}$. The points P and Q lie on the circumference of the circle.

(i) Find the exact area of the sector OPQ .

1

(ii) Find the exact area of $\triangle OPQ$.

1

(iii) Hence find the exact area of the shaded minor segment.

1

(c) Find the equation of the tangent to the curve $y = e^{3x+1}$ at the point $(0, e)$.

2

(d) Solve $\cos x = -\frac{\sqrt{3}}{2}$, for $0 \leq x \leq 2\pi$.

2

QUESTION ELEVEN (12 marks) Use a separate writing booklet.

Marks

- (a) Sketch graphs of the following functions on separate number planes, showing all intercepts with the axes and other important features.

(i) $y = 2^x - 4$

2

(ii) $y = \frac{1}{x-3}$

2

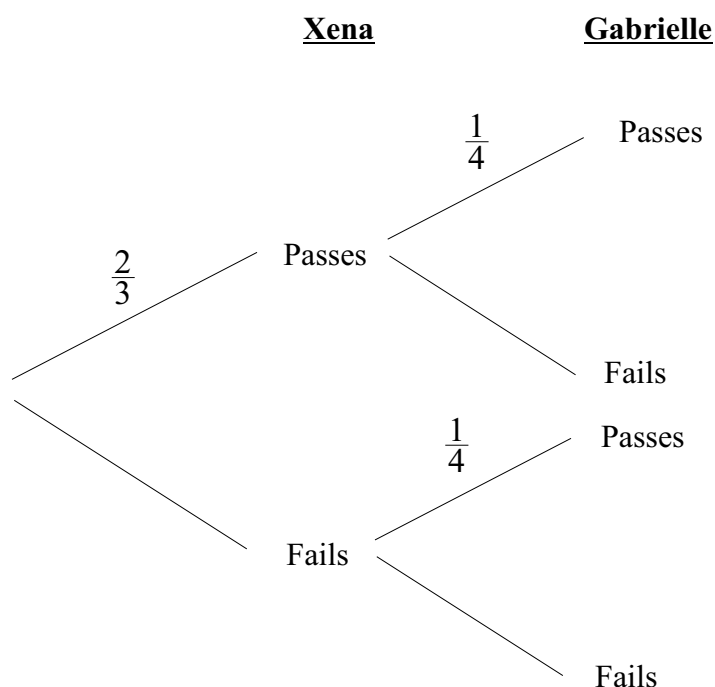
(iii) $y = \log_3 x$

2

(iv) $y = x(x-1)(x+1)(x+3)$

2

- (b) Xena and Gabrielle are going to sit their driving test. The probability that Xena passes her driving test is $\frac{2}{3}$. The probability that Gabrielle passes her driving test is $\frac{1}{4}$. Part of the probability tree diagram for this scenario is shown below.



- (i) Copy and complete the tree diagram showing the probabilities on each branch.
- (ii) Write a list of possible outcomes.
- (iii) What is the probability that only one of the girls passes their driving test?

1

1

2

QUESTION TWELVE (12 marks) Use a separate writing booklet.**Marks**

- (a) If $\sin \theta = -\frac{2}{5}$ and $\cos \theta > 0$, find the exact value of $\tan \theta$.

2

- (b) Consider the function $f(x) = x^2 + 7x - 10$.

(i) Write down an expression for $f(x + h)$.

1

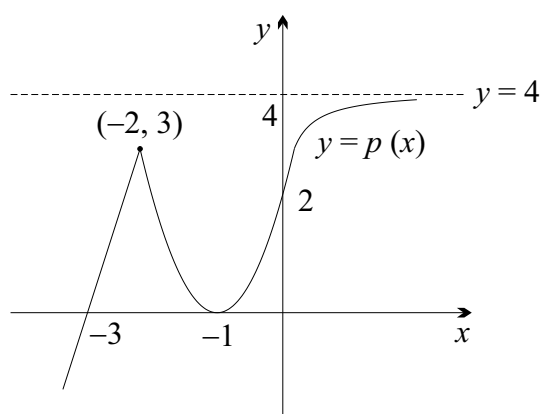
(ii) Differentiate $f(x)$ from first principles to show that $f'(x) = 2x + 7$.

2

(iii) What is the gradient of the normal to $y = f(x)$ at the point $(1, -2)$?

1

(c)

2

The diagram above shows an unknown function $y = p(x)$.

The function $y = p(x)$ is transformed to give $y = -p(-x)$. Sketch $y = -p(-x)$ showing any intercepts with the axes and other important features.

- (d) If $x = \log_a 3$, $y = \log_a 5$ and $z = \log_a 7$, write the expression $\log_a \left(\frac{75}{49a} \right)$ in terms of x , y and z only.

2

- (e) Determine whether the function $f(x) = \frac{x^3 + 3x}{x^4 + x^2 - 1}$ is even, odd or neither. You must show all working.

2

QUESTION THIRTEEN (12 marks) Use a separate writing booklet.

Marks

- (a) On any given day in Sydney, the probability that it is raining is $\frac{1}{3}$, the probability that there is heavy traffic is $\frac{7}{12}$ and the probability that there is heavy traffic or it is raining is $\frac{2}{3}$.

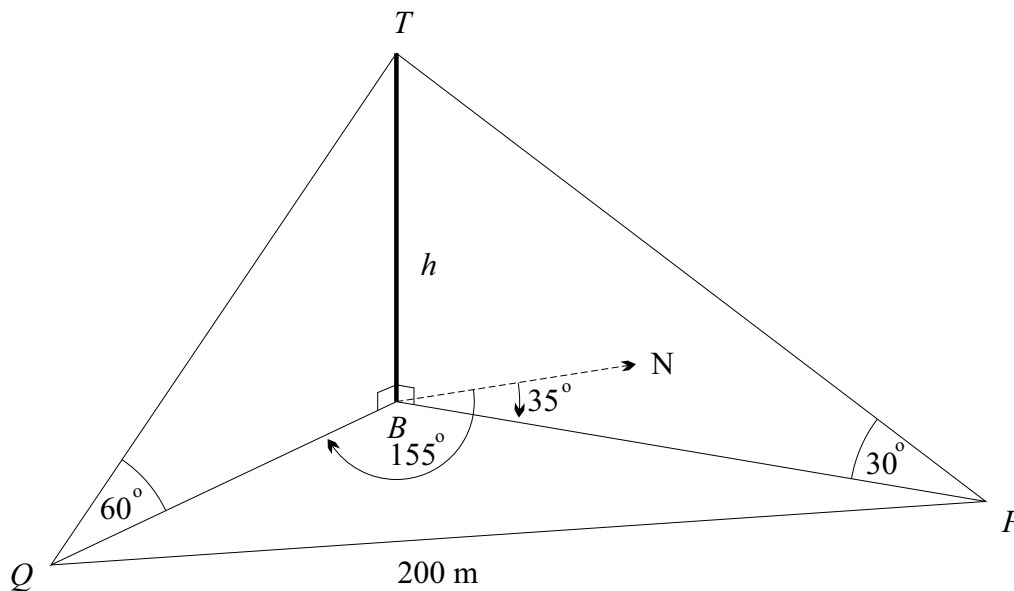
(i) Find the probability that it is raining and there is heavy traffic.

1

(ii) Find the probability that there is heavy traffic given that it is raining.

2

(b)



A vertical tower is observed from two landmarks P and Q on level ground. From P , the angle of elevation to the top of the tower T is 30° . From Q , the angle of elevation to the top of the tower T is 60° . From the base of the tower B , P is on a bearing of 035°T and Q is on a bearing of 155°T . It is known that P and Q are 200 m apart. Let h represent the height of the tower BT .

(i) Explain why $\angle PBQ = 120^\circ$.

1

(ii) Show that $BP = \sqrt{3}h$ and write a similar expression for BQ .

2

(iii) Show that $\frac{13h^2}{3} = 200^2$ and hence find the height of the tower h . Give your answer correct to the nearest metre.

2

QUESTION THIRTEEN (Continued)

- (c) In January 2018, a farmer accidentally spread a dangerous chemical on a paddock. The concentration of the chemical in the soil was initially measured to be 7 kg/ha. One year later, the concentration was found to be 3.2 kg/ha.

It is known that the concentration C of the chemical in kg/ha is given by the formula $C = C_0 e^{-kt}$ where t is the time in years after the chemical was spread and C_0 is a constant.

- (i) Find the values of C_0 and k , giving your answers in exact form. 2
- (ii) It is safe to use the paddock when the concentration falls below 0.2 kg/ha. During which year will it first become safe for the farmer to use the paddock again? 2

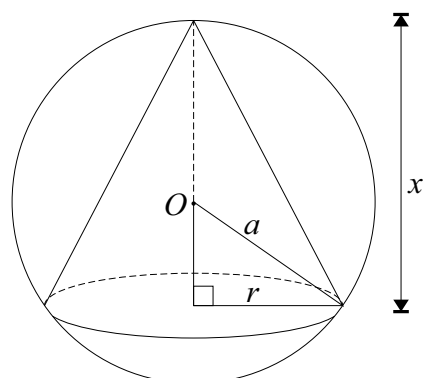
QUESTION FOURTEEN (12 marks) Use a separate writing booklet.

Marks

- (a) Prove that $\frac{(1 - \sin \theta)(1 + \sin \theta)}{\cos^2 \theta} = 1$.

2

(b)



The diagram above shows a cone inscribed in a sphere. The sphere has radius a and centre O . The cone has a base of radius r and height x .

- (i) Show that $r^2 = 2ax - x^2$.

1

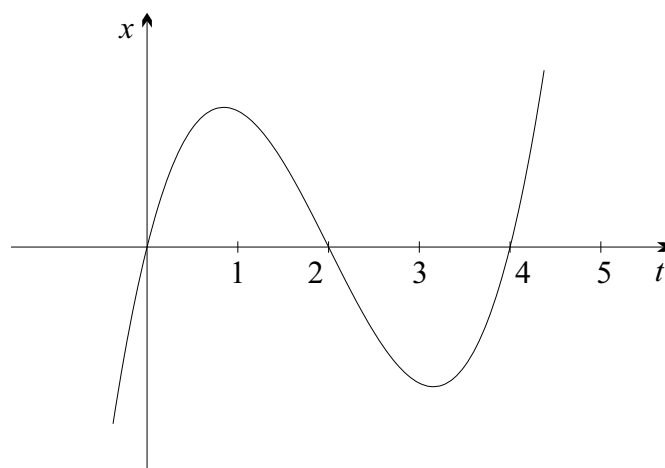
- (ii) The volume of a cone is given by the formula $V = \frac{1}{3}Ah$ where A is the area of the base and h is the perpendicular height of the cone. Show that the volume of this cone is given by $V = \frac{1}{3}\pi(2ax^2 - x^3)$.

1

- (iii) Find the value of x so that $\frac{dV}{dx} = 0$, given $0 < x < 2a$.

2

(c)



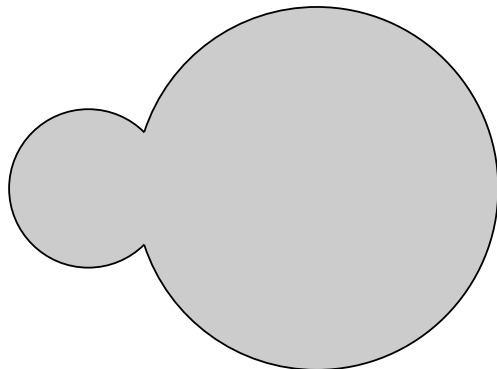
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The graph above shows the function $x = f(t)$.

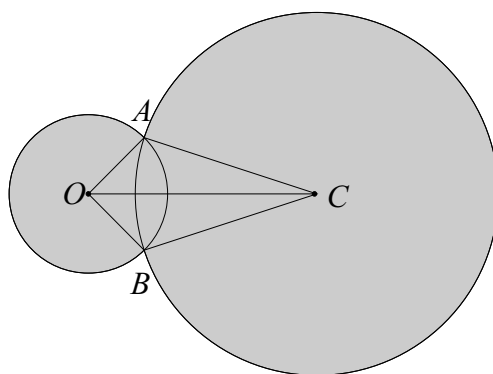
Sketch a possible graph of $\frac{dx}{dt}$ as a function of t .

QUESTION FOURTEEN (Continued)

(d)



An advertising logo is formed from two circles which intersect as shown in the diagram above.



The circles intersect at A and B and have centres at O and C . The diagram above shows the radii OA , OB , CA and CB of each circle. The radius of the circle centred at O is 0.5 m and the radius of the circle centred at C is $\frac{\sqrt{3}}{2}$ m. The length of OC is 1 m.

- (i) Show that $\angle OAC = \frac{\pi}{2}$.
- (ii) Find the area of the quadrilateral $AOBC$.
- (iii) Find the total area of the logo. Give your answer in exact form.

1

1

2

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

B L A N K P A G E

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2019
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MATHEMATICS ADVANCED
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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Mathematics Advanced Form V Solutions

Question 1

$$\begin{aligned} & (3\sqrt{5} - 5)^2 \\ &= 45 - 30\sqrt{5} + 25 \\ &= 70 - 30\sqrt{5} \end{aligned}$$

(B)

Question 2

$$\begin{aligned} 14x + 4y &= 8 \\ 5x + 4y &= 404 \end{aligned}$$

(B)

Question 3

$$\log_a 2m - \log_a m$$

$$= \log_a \frac{2m}{m}$$

$$= \log_a 2$$

(D)

Question 4

$$\frac{d}{dx} e^{2x} = 2e^{2x}$$

(B)

Question 5

$$A = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 3 \times 5 \sin 94$$

(C)

Question 6

$$4 - x^2 > 0$$

$$(2-x)(2+x) > 0 \quad \cancel{-2} \quad \cancel{2}$$

$$-2 < x < 2$$

(C)

Question 7

$$\text{either } P(A|B) = P(A)$$

$$\text{or } P(A \cap B) = P(A) \times P(B)$$

$$A \times$$

$$B \rightarrow P(A \cap B) = P(A) \times P(B)$$

$$C \rightarrow X$$

$$D \times$$

Answer (B)

Question 8

$$\begin{aligned} f(g(x)) &= |x|^2 + 3|x| - 4 \\ &= x^2 + 3|x| - 4 \end{aligned}$$

$$= \begin{cases} x^2 + 3x - 4 & x \geq 0 \\ x^2 - 3x - 4 & x < 0 \end{cases}$$

(A)

Question 9

a) $f(x) = x^2 - 2x + 3$

i) $f(2) = 2^2 - 2 \times 2 + 3$
 $= 3$ ✓

ii) $f(a+2) = (a+2)^2 - 2(a+2) + 3$
 $= a^2 + 4a + 4 - 2a - 4 + 3$
 $= a^2 + 2a + 3$ ✓

ii) $f(a) + 2 = a^2 - 2a + 3 + 2$
 $= a^2 - 2a + 5$ ✓

b) one-to-many ✓

c) $\frac{2(\sqrt{5}+2)}{(\sqrt{5}-2)(\sqrt{5}+2)}$

$$= \frac{2(\sqrt{5}+2)}{5-4}$$

$$= 2\sqrt{5} + 4 \quad \text{or} \quad 2(\sqrt{5}+2) \quad \checkmark$$

d) $|2x - 1| = 11$

$$2x - 1 = 11$$

$$2x = 12$$

$$x = 6$$

$$2x - 1 = -11$$

$$2x = -10$$

$$x = -5$$

$$x = 6 \quad \text{or} \quad x = -5$$

✓ ✓

e) Domain: $-3 \leq x \leq 3$ ✓
Range: $0 \leq y \leq 3$ ✓

f) $P(RR) = \frac{2}{5} \times \frac{1}{4}$
 $= \frac{1}{10}$ ✓

g) i) $\sin \frac{2\pi}{3}$
 $= \sin \pi/3$
 $= \frac{\sqrt{3}}{2}$ ✓

ii) $\cos \frac{5\pi}{4}$
 $= -\cos \pi/4$
 $= -\frac{1}{\sqrt{2}}$ ✓

Question 10

$$\text{a) i) } y = 2x^3 \\ \frac{dy}{dx} = 6x^2 \quad \checkmark$$

$$\text{ii) } y = x^{-2} \\ \frac{dy}{dx} = -2x^{-3} \quad \checkmark \\ = -\frac{2}{x^3}$$

$$\text{iii) } y = (3x+2)^4 \\ \frac{dy}{dx} = 4(3x+2)^3 \times 3 \quad \checkmark \\ = 12(3x+2)^3$$

$$\text{iv) } u = x^3 \quad v = 3x+2 \\ u' = 3x^2 \quad v' = 3 \\ v^2 = (3x+2)^2$$

$$\frac{dy}{dx} = \frac{3x^2(3x+2) - 3x^3}{(3x+2)^2} \quad \checkmark$$

$$= \frac{9x^3 + 6x^2 - 3x^3}{(3x+2)^2}$$

$$= \frac{6x^3 + 6x^2}{(3x+2)^2} \quad \checkmark$$

$$\begin{aligned}
 b) \ i) \ A &= \frac{r^2 \theta}{2} \\
 &= \frac{5^2 \times \pi/4}{2} \\
 &= \frac{25\pi}{8} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 ii) \ A &= \frac{r^2 \sin \theta}{2} \\
 &= \frac{5^2 \times \sin \pi/4}{2} \\
 &= \frac{25}{2} \times \frac{1}{\sqrt{2}} \\
 &= \frac{25\sqrt{2}}{4} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 iii) \ A &= \frac{25\pi}{8} - \frac{25\sqrt{2}}{4} \quad \checkmark \\
 &= \frac{25\pi - 50\sqrt{2}}{8}
 \end{aligned}$$

$$\begin{aligned}
 c) \quad y &= e^{3x+1} & (0, e) \\
 \frac{dy}{dx} &= 3e^{3x+1} \\
 m_T &= 3e^{0+1} \\
 &= 3e \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 y - e &= 3e(x - 0) \quad \checkmark \\
 y &= 3ex + e
 \end{aligned}$$

$$d) \quad \cos x = -\frac{\sqrt{3}}{2}$$

$$\cos \alpha = \frac{\sqrt{3}}{2} \quad \alpha \text{ is acute}$$

$$\alpha = \frac{\pi}{6} \quad \checkmark \quad \text{p. 2+3}$$

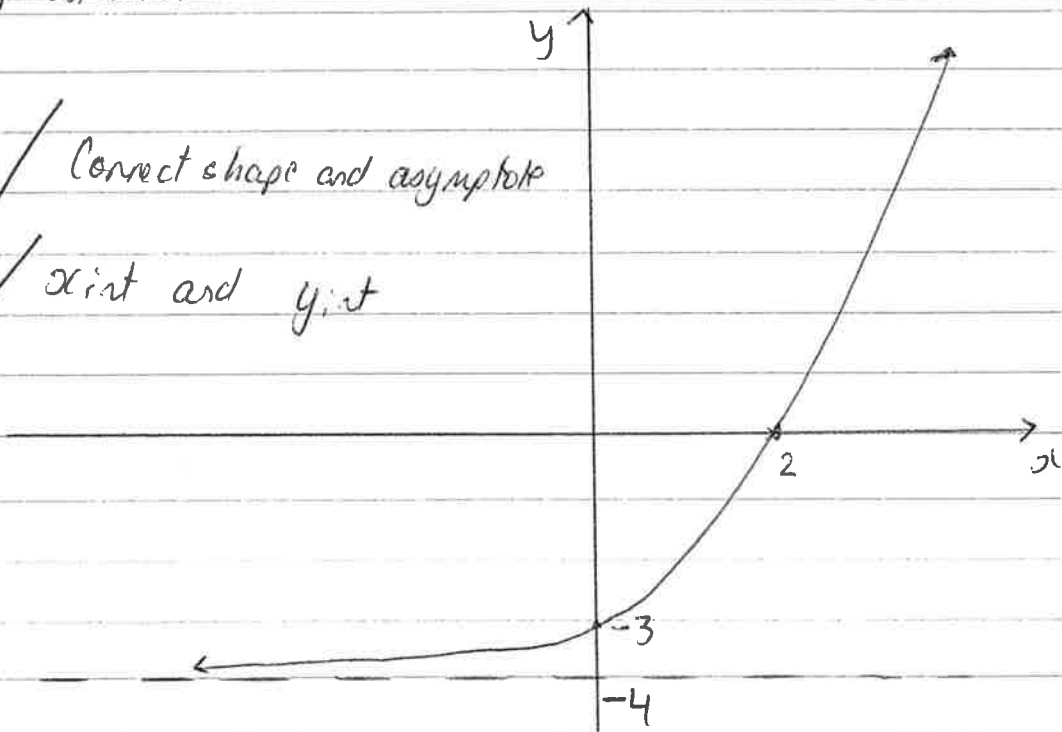
$$x = \pi - \frac{\pi}{6} \text{ or } \pi + \frac{\pi}{6}$$

$$= \frac{5\pi}{6} \text{ or } \frac{7\pi}{6} \quad \checkmark$$

Question 11

- ✓ Correct shape and asymptote
- ✓ x-int and y-int

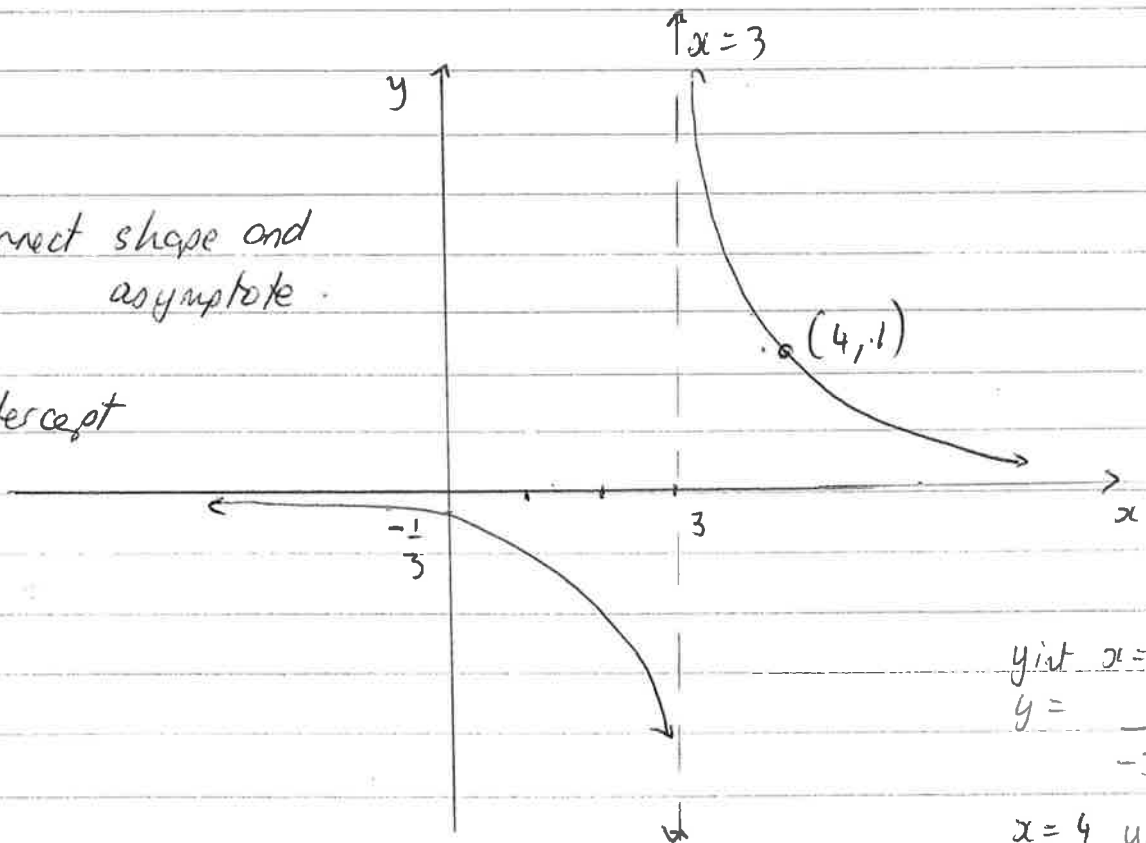
$$\begin{aligned} x &= 0 \\ 2^x &= 4 \\ x &= 2 \end{aligned}$$



$$y = -4$$

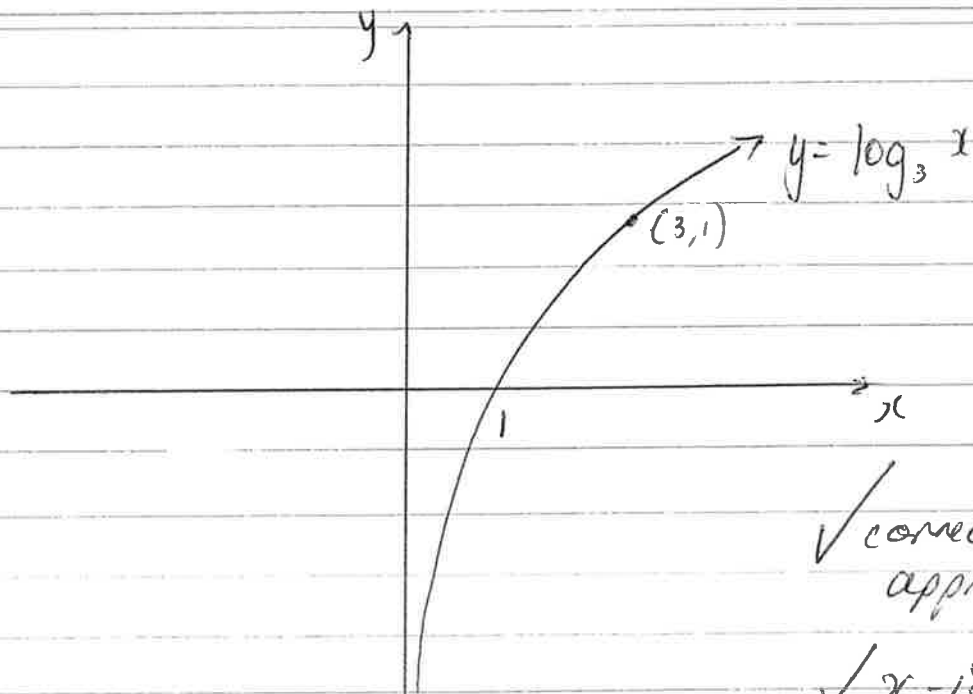
ii)

- ✓ correct shape and asymptote
- ✓ y-intercept



$$\begin{aligned} \text{y-int } x &= 0 \\ y &= \frac{1}{-3} \\ x &= 4 \quad y &= 1 \end{aligned}$$

iii)



✓ correct shape and
approaching y-axis

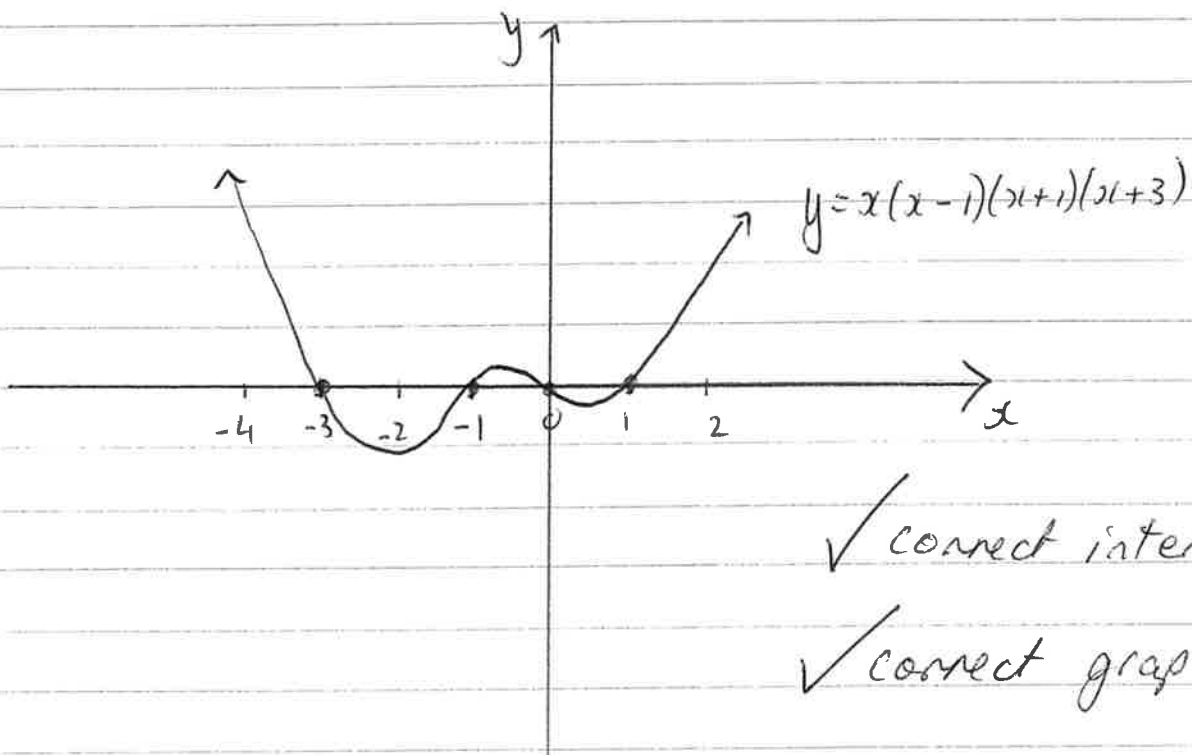
✓ x-intercept

iv)

x	-4	-3	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y	+	0	-	0	+	0	-	0	+

x-int $x = 0, 1, -1, -3$

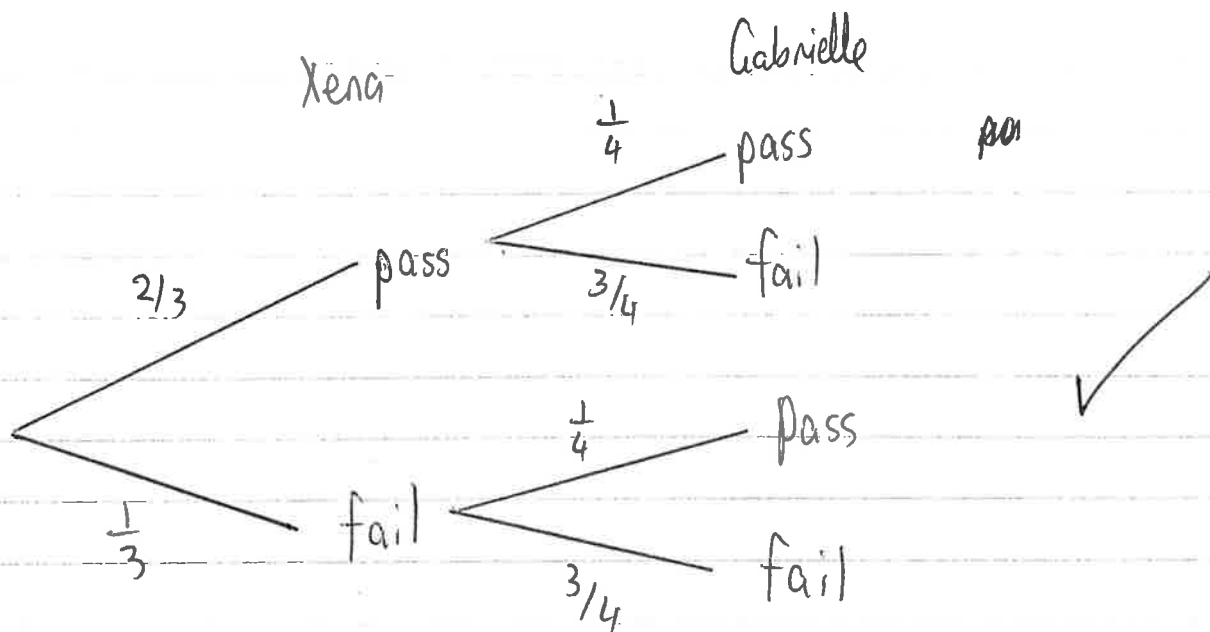
table of signs



✓ correct intercepts

✓ correct graph

b) i)



ii)

Xena	Gabrielle
P	P
P	F
F	P
F	F



iii) $P(\text{only one fails}) = P(PF) + P(FP)$

$$= \frac{2}{3} \times \frac{3}{4} + \frac{1}{3} \times \frac{1}{4}$$

$$= \frac{7}{12}$$

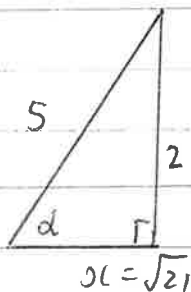


Question 12

a) $\sin \theta = -\frac{2}{5}$ $\cos \theta > 0$

Quadrant 4

let $\sin \alpha = \frac{2}{5}$ α acute



$$\begin{aligned}x^2 &= 5^2 - 2^2 \quad (\text{Pythagoras' Theorem}) \\x^2 &= 21 \\x &= \sqrt{21}\end{aligned}$$

$$\begin{aligned}\tan \theta &= -\tan \alpha \\&= -\frac{2}{\sqrt{21}} \quad \checkmark \\&= -\frac{2\sqrt{21}}{21}\end{aligned}$$

b) i) $f(x) = x^2 + 7x - 10$ $\checkmark$

$$\begin{aligned}f(x+h) &= (x+h)^2 + 7(x+h) - 10 \\&= x^2 + 2xh + h^2 + 7x + 7h - 10\end{aligned}$$

ii) $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 7x + 7h - 10 - x^2 - 7x + 10}{h} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 7h}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h + 7$$



or showing $\frac{h(2x+h+7)}{h}$
factorised

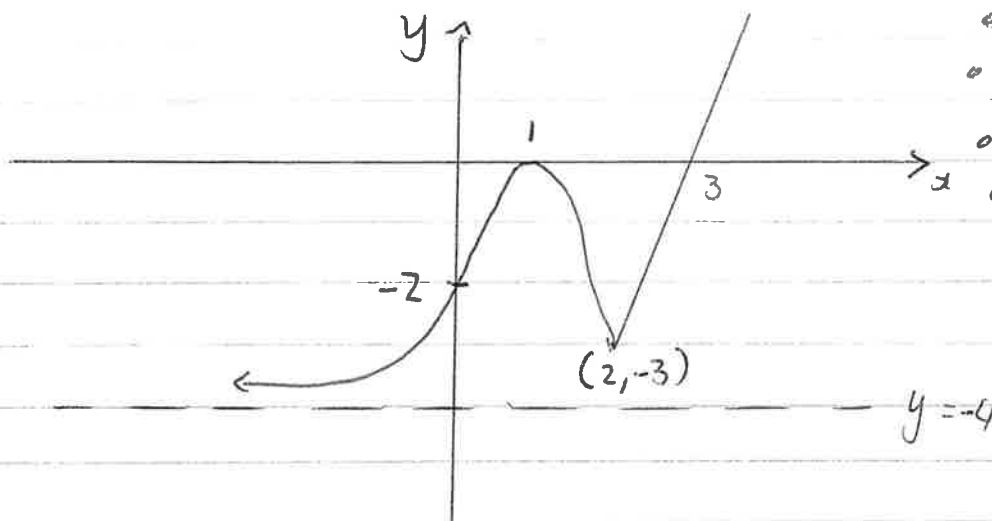
$$= 2x + 7$$

$$\text{iii) } M_T = 2(1) + 7 \\ = 9$$

$$m_N = -\frac{1}{9}$$



c)



- x-intercepts
- y-intercepts
- sharp point
- correct orientation
- asymptote clearly shown.

$$y = -4$$

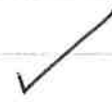
✓ 2 of 5 features

✓ all 5 features

$$\text{d) } \log_a \left(\frac{75}{49a} \right)$$

$$= \log_a 75 - \log_a 49a$$

$$= \log_a 3 + 2\log_a 5 - (\log_a 7^2 + \log_a a)$$



$$= \log_a 3 + 2\log_a 5 - 2\log_a 7 - 1$$

$$= x + 2y - 2z - 1$$



$$e) \quad f(x) = \frac{x^3 + 3x}{x^4 + x^2 - 1}$$

$$\begin{aligned} f(-x) &= \frac{(-x)^3 + 3(-x)}{(-x)^4 + (-x)^2 - 1} \\ &= \frac{-x^3 - 3x}{x^4 + x^2 - 1} \end{aligned}$$

✓ showing
not even

$\neq f(x)$ So the function is not even

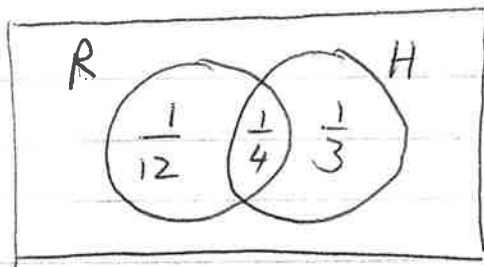
$$\begin{aligned} -f(-x) &= -\left(\frac{-x^3 - 3x}{x^4 + x^2 - 1} \right) \\ &= \frac{x^3 + 3x}{x^4 + x^2 - 1} \\ &= f(x) \end{aligned}$$

✓ Must explicitly show
either multiplying $f(-x)$ by
-1 or multiplying
 $f(x)$ by -1.
Showing odd

Since $f(x) = -f(-x)$ the function is odd.

Question 13

a) i)



$$P(R \cap H) = \frac{1}{3} + \frac{7}{12} - \frac{2}{3}$$

$$= \frac{1}{4} \quad \checkmark$$

$$ii) P(H|R) = \frac{P(H \cap R)}{P(R)} \quad \checkmark$$

$$= \frac{\frac{1}{4}}{\frac{1}{3}}$$

$$= \frac{3}{4} \quad \checkmark$$

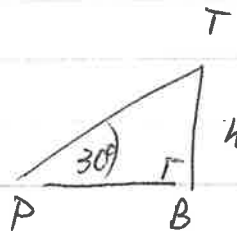
$$b) i) \angle PBQ = 155^\circ - 35^\circ \text{ (adjacent angles)}$$

$$\angle PBQ = 120^\circ$$

$$ii) \tan 30^\circ = \frac{h}{BP} \quad \checkmark$$

$$BP \times \frac{1}{\sqrt{3}} = h$$

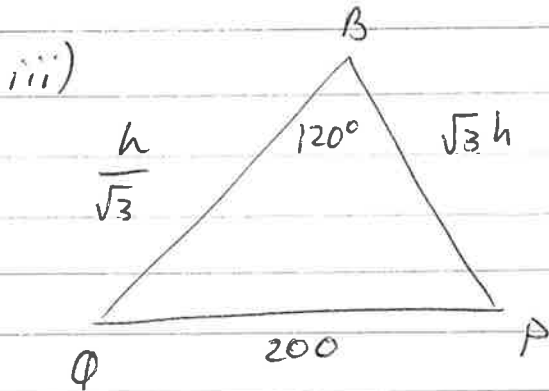
$$BP = \sqrt{3}h \text{ as required}$$



$$\tan 60^\circ = \frac{h}{BQ}$$

$$\sqrt{3} \times BQ = h$$

$$BQ = \frac{h}{\sqrt{3}}$$



$$200^2 = (\sqrt{3}h)^2 + \left(\frac{h}{\sqrt{3}}\right)^2 - 2 \times \sqrt{3}h \times \frac{h}{\sqrt{3}} \cos 120^\circ$$



$$200^2 = 3h^2 + \frac{h^2}{3} - 2h^2(-\cos 60)$$

$$200^2 = 3h^2 + \frac{h^2}{3} + h^2$$

$$200^2 = \frac{13h^2}{3} \quad \text{as required}$$

$$13h^2 = 120000$$

$$h^2 = \frac{120000}{13}$$

$$h \approx 96.07689 \dots$$

$$h \approx 96 \text{ m (nearest metre)}$$



Question 13

c) i) $C = C_0 e^{-kt}$

$t=0 \quad C=7 \quad \text{so} \quad C_0=7 \quad \checkmark$

$$C = 7e^{-kt}$$

$t=1 \quad C=3.2$

$$3.2 = 7e^{-k}$$

$$\frac{16}{35} = e^{-k}$$

$$-k = \log_e \frac{16}{35}$$

$$k = -\log_e \frac{16}{35} \quad \text{or} \quad k = \log_e \frac{35}{16} \quad \checkmark$$

ii) $C = 7e^{\log_e \frac{16}{35} \times t}$

We want $C=0.2$

$$0.2 = 7e^{\log_e \frac{16}{35} \times t}$$

$$\frac{1}{35} = e^{\log_e \frac{16}{35} \times t}$$

$$\log_e \frac{1}{35} = \log_e \frac{16}{35} \times t$$

$$t = \frac{\log_e \frac{1}{35}}{\log_e \frac{16}{35}}$$

$\checkmark$ substitute & make progress

$$t \approx 4.54207 \dots$$

During the year 2022

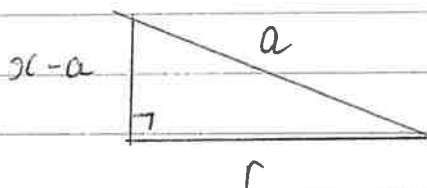
$$\text{Answer } 2018.74$$

$\checkmark$

Question 14

$$\begin{aligned} \text{a) LHS} &= \frac{(1 - \sin \theta)(1 + \sin \theta)}{\cos^2 \theta} \\ &= \frac{1 - \sin^2 \theta}{\cos^2 \theta} \quad \checkmark \\ &= \frac{\cos^2 \theta}{\cos^2 \theta} \quad \checkmark \\ &= 1 \\ &= \text{RHS as required} \end{aligned}$$

b) i)



$$r^2 + (x-a)^2 = a^2 \quad \checkmark \quad (\text{pythagoras' theorem})$$

$$r^2 + x^2 - 2ax + a^2 = a^2$$

$$r^2 = 2ax - x^2 \quad \text{as required}$$

$$\begin{aligned} \text{ii) } A &= \pi r^2 \\ &= \pi (2ax - x^2) \end{aligned}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi (2ax - x^2) x \quad \checkmark$$

$$= \frac{1}{3} \pi (2ax^2 - x^3) \quad \text{as required}$$

$$\text{iii)} \quad \frac{dV}{dx} = \frac{1}{3} \pi (4ax - 3x^2) \quad \checkmark$$

Solve $\frac{dV}{dx} = 0$

$$\frac{1}{3} \pi (4ax - 3x^2) = 0$$

$$4ax - 3x^2 = 0$$

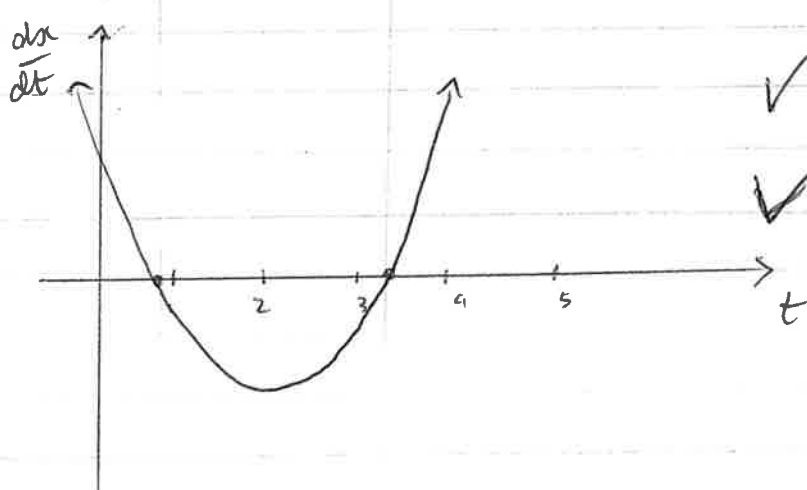
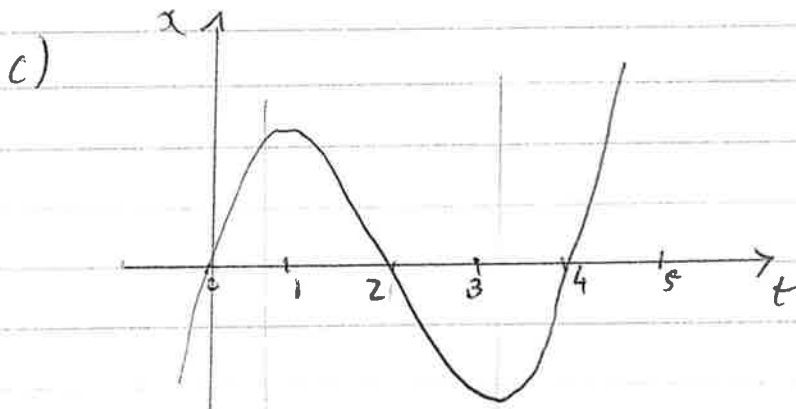
$$x(4a - 3x) = 0$$

$$x = 0 \quad \text{or} \quad 3x = 4a$$

$$x = \frac{4a}{3} \quad \checkmark$$

but $0 < x < 2a$

So $x = \frac{4a}{3}$ only.



✓ shape

✓ intercepts:

(note x -ints are slightly off $x=1$ & $x=3$)

$$\begin{aligned}
 d) \quad i) \quad OA^2 + AC^2 &= 0.5^2 + \left(\frac{\sqrt{3}}{2}\right)^2 \\
 &= \frac{1}{4} + \frac{3}{4} \quad \checkmark \\
 &= 1 \\
 &= OC^2
 \end{aligned}$$

Since pythagoras theorem holds, $\angle OAC$ is 90°

$$\begin{aligned}
 ii) \quad \text{Area} &= 2 \times \frac{1}{2} \times OA \times AC \\
 &= OA \times AC \\
 &= \frac{1}{2} \times \frac{\sqrt{3}}{2} \\
 &= \frac{\sqrt{3}}{4} \text{ m}^2 \quad \checkmark
 \end{aligned}$$

iii) let $\angle OAC = \alpha$ and $\angle ACO = \theta$

$$\sin \alpha = \frac{\sqrt{3}}{2}$$

$$\alpha = \frac{\pi}{3} \quad \text{So } \theta = \frac{\pi}{6} \quad (\text{angle sum } \triangle AOC)$$

$$\text{So } \angle AOB = 2\alpha = \frac{2\pi}{3} \quad \text{and } \angle ACB = \frac{\pi}{3}$$

$$\text{Reflex } \angle AOB = \frac{4\pi}{3} \quad \text{and reflex } \angle ACB = \frac{5\pi}{3}$$

$$\text{Area Sector AOB} = \frac{r^2 \theta}{2}$$

$$= \left(\frac{1}{2}\right)^2 \times \frac{4\pi}{3} \times \frac{1}{2}$$

$$= \frac{1}{4} \times \frac{4\pi}{3} \times \frac{1}{2}$$

$$= \frac{\pi}{6}$$

✓ Finding at least one reflex angle and correct area of sector

$$\text{Area sector ACB} = \left(\frac{\sqrt{3}}{2}\right)^2 \times \frac{5\pi}{3} \times \frac{1}{2}$$

$$= \frac{3}{4} \times \frac{5\pi}{3} \times \frac{1}{2}$$

$$= \frac{5\pi}{8}$$

$$\text{Total Area} = \frac{\pi}{6} + \frac{5\pi}{8} + \frac{\sqrt{3}}{4}$$

$$= \frac{4\pi + 15\pi + 6\sqrt{3}}{24}$$

$$= \frac{19\pi + 6\sqrt{3}}{24}$$

✓